TripletMarginLoss#

https://pytorch.org/docs/stable/generated/torch.nn.TripletMarginLoss.html#to

Description:

Creates a criterion that measures the triplet loss given an input tensors x1x1x1, x2x2x2, x3x3x3 and a margin with a value greater than 000. This is used for measuring a relative similarity between samples. A triplet is composed by a, p and n (i.e., anchor, positive examples and negative examples respectively). The shapes of all input tensors should be (N,D)(N, D)(N,D). The distance swap is described in detail in the paper Learning shallow convolutional feature descriptors with triplet losses by V. Balntas, E. Riba et al. The loss function for each sample in the mini-batch is: The norm is calculated using the specified p value and a small constant ε\varepsilonε is added for numerical stability.

Function Signature

```
class torch.nn.TripletMarginLoss(margin=1.0, p=2.0, eps=1e-06, swap=False, size_average=None, reduce=None, reduction='mean')[source]#

Parameters

Shape:

forward(anchor, positive, negative)[source]#

Return type
```

Returns:

Tensor

Code Examples

```
>>> triplet_loss = nn.TripletMarginLoss(margin=1.0, p=2, eps=1e-7)
>>> anchor = torch.randn(100, 128, requires_grad=True)
>>> positive = torch.randn(100, 128, requires_grad=True)
>>> negative = torch.randn(100, 128, requires_grad=True)
>>> output = triplet_loss(anchor, positive, negative)
>>> output.backward()
```

Detailed Documentation

Documentation

Creates a criterion that measures the triplet loss given an input tensors x1x1x1, x2x2x2, x3x3x3 and a margin with a value greater than 000. This is used for measuring a relative similarity between samples. A triplet is composed by a, p and n (i.e., anchor, positive examples and negative examples respectively). The shapes of all input tensors should be (N,D)(N, D)(N,D).

The distance swap is described in detail in the paper Learning shallow convolutional feature descriptors with triplet losses by V. Balntas, E. Riba et al.

The loss function for each sample in the mini-batch is:

The norm is calculated using the specified p value and a small constant ε\varepsilonε is added for numerical stability.

See also TripletMarginWithDistanceLoss, which computes the triplet margin loss for input tensors using a custom distance function.

margin (float, optional) – Default: 111.

p (int, optional) – The norm degree for pairwise distance. Default: 222.

eps (float, optional) – Small constant for numerical stability. Default: 1e−61e-61e−6.

swap (bool, optional) – The distance swap is described in detail in the paper Learning shallow convolutional feature descriptors with triplet losses by V. Balntas, E. Riba et al. Default: False.

size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: (N,D)(N, D)(N,D) or (D)(D)(D) where DDD is the vector dimension.

Output: A Tensor of shape (N)(N)(N) if reduction is 'none' and input shape is (N,D)(N, D)(N,D); a scalar otherwise.

Runs the forward pass.